DAVID KURNIAWAN

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Education

Xiamen University Malaysia Expected Sep 2028
B.Eng. (Hons.) in Artificial Intelligence GPA: 3.84/4.00
Dean's List: 3 consecutive semesters; Top 16% of cohort
Relevant Coursework: Data Structures, Object-Oriented Programming (C++), Applied Machine Learning, Deep Learning, Natural Language Processing

Technical Skills

Programming: Python, JavaScript, TypeScript, C, C++, HTML, CSS
Backend & Frontend: FastAPI, Flask, Node.js, React, REST APIs, WebSockets
Databases: PostgreSQL, Supabase, pgvector, Firebase, Qdrant
Cloud & Infrastructure: Docker, Cloudflare Workers, Railway, Vercel, Cloudflare DNS, Resend, CI/CD (GitHub Actions)
Engineering Practices: System design, authentication & authorisation, database schema design & migrations (Alembic), unit & integration testing, logging, error handling, model evaluation
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, OpenCV, LangChain, LangGraph, Transformers (Hugging Face), FAISS, BM25, Weights & Biases
Languages: Indonesian (Native), English (IELTS 7.0), Mandarin Chinese (Elementary)

Selected Projects

KerjaCerdas — AI Job-Matching Platform 2026
Lead Engineer / Backend | FastAPI, LangGraph, PostgreSQL/pgvector, Docker, React 18

- Built an AI-powered job-matching platform for Indonesian job seekers and SMEs using FastAPI, LangGraph, PostgreSQL/pgvector, and React; implemented GitHub Actions CI, vector-search integration tests, and Alembic migration checks.
- Engineered an explainable candidate-ranking engine using 768-dimensional Gemini embeddings, pgvector HNSW search, and a weighted composite score across skill, experience, education, and semantic similarity; introduced three-tier skill evidence levels (claimed, quiz-verified, employer-confirmed).
- Added pre-inference CV PII redaction and explicit consent prompts to reduce unnecessary exposure of personal data; implemented model fallback chains and token-efficiency checks to manage inference reliability and API usage.

Orion — Automated Expense-Claim Workflow 2026
Tech Lead / Backend, 4-person team | FastAPI, LangGraph, Pydantic, LangSmith

- Built a six-stage expense-claim automation workflow using FastAPI and LangGraph, integrating PDF extraction, deterministic policy validation, fuzzy duplicate detection, approval routing, and audit logging.
- Built a deterministic policy-enforcement engine with fuzzy duplicate detection, PDF extraction, and structured audit logging; enforced strict data contracts between stages using Pydantic schemas.
- - Integrated LangSmith to trace workflow executions and monitor LLM token usage and costs; configured GitHub Actions to enforce an 85% code-coverage threshold and run nightly regression tests against live LLM APIs.

Indonesian Legal RAG 2026
LangChain, FAISS, BM25, Hugging Face, Weights & Biases

- Built a legal RAG pipeline combining BM25 sparse retrieval, FAISS dense search, HyDE query expansion, and cross-encoder reranking to retrieve relevant Indonesian labour-law provisions; added a web-search fallback for low-confidence queries.
- Fine-tuned Llama 3 8B on a curated legal instruction dataset using supervised fine-tuning; tracked experiments with Weights Biases and published the resulting model to Hugging Face.

Awards & Achievements

Top 41 / 2,000+ Teams — PIDI × Digdaya Hackathon (Tier 3) Mar – Sep 2026
Led engineering for KerjaCerdas, overseeing system architecture, AI inference, and full-stack implementation.
Silver Award — SEA-CICSIC 2026 International Research Competition Jun 2026
Led AI strategy and developed the proposal and pitch deck for Omni-QC, an AI-based manufacturing defect-prediction and quality-control concept.
3rd Place — DPickleball AI Competition Oct 2025
Led a 3-person team developing a reinforcement-learning agent using RecurrentPPO, LSTM, CNN, and Unity ML-Agents; implemented reward shaping and self-play league training.
Top 20% Globally — International Quant Championship, Stage 1 Apr 2025